

Problem 3.8

If $A(4) = 1200$, $A(n) = 1800$, and $i = 0.06$.

- (a) What is $A(0)$ assuming simple interest?
- (b) What is n ?

Solution

For simple interest, the accumulation function is

$$a(t) = 1 + it.$$

- (a) Since

$$A(4) = 1200 = x(1 + 0.06(4)),$$

we get

$$1200 = x(1.24).$$

Therefore,

$$x = \frac{1200}{1.24} = 967.7419.$$

Hence,

$$\boxed{A(0) = 967.7419.}$$

- (b) Now use

$$A(n) = 1800 = 967.7419(1 + 0.06n).$$

Then

$$\frac{1800}{967.7419} = 1 + 0.06n,$$

so

$$n = \frac{\frac{1800}{967.7419} - 1}{0.06}.$$

Therefore,

$$n \approx 14.33.$$